

Wanja Wiese ~ Active Inference Insights 016 ~ Consciousness, Information

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hello everyone and welcome back to
active inference insights I'm your host
darus parv Wayne and today I'm delighted
to be speaking to Vana Visa van is a
postdoctoral fellow at Johannes
Gutenberg University mines in the
department of Ro University B sorry the
internet has lied to me the internet has
lied to me I don't like this
introduction bit already um but I think
this is true nonetheless his research
centers on Consciousness medal rep
presentation philosophy of cognitive
science philosophy of mind and of course
Act of inference uh we also both share a
love of self- modeling attention
autopoietic intentionality and much much
more and I'm sure these are just some of
the topics that we will discuss today V
welcome to the show thank you so much
for joining me thank you for having me
it's a pleasure this is really exciting
uh yeah as I said in the intro we share
a lot of interests uh that might appear
quite Niche to people but actually you
know as far as I can tell pertaining to
their lived experience every every
moment so it's it's worthwhile
dissecting funnily enough the first time
I think I'd heard of you was actually
from jackob Javi um in the conversation
I had with him and he said that you know
Thomas metzinger's work perhaps better
than Thomas and certainly better than
anyone else um so so I I wonder

whether we could start by outlining some of these crucial um Concepts that sometimes seem a bit counterintuitive to people so for example when I say we both share a love of self- modeling or phenomenal self modeling what does that mean what is this sort of acronym PSM the phenomenal self modeling yeah good so um maybe as a um disclaimer of course um Thomas knows more about this than me he's um uh in invented the term self models and has developed the self model theory of subjectivity over many many years and um actually I I I think I can say that there's probably no living philosopher who has read as much relevant literature about topic as as Thomas because he's devoted his entire career to this and works a lot and of of course I would never um even even if I tried be in a position to know as much about this as as he does and I I think it might be a good idea even to invite him to this podcast he's been invited he has said yes that was a while ago I've tried to push it so Thomas if you're listening which I think he might be you know you never know um please do come on because I do yeah so actually it's quite likely that some of the things that I've been saying might not be 100% true so Thomas if you're listening now you may want to correct some of the misrepresentations of your work um but anyway yeah what what is the self model so for me it's I think easiest to First think about bodily self models so um just imagine

some creature or a robot that has a body and has to control the body and as we all know in many cases control can be facilitated or improved if you have a model of the thing that you're controlling and if you want to control your body and the interaction with the environment it can be highly useful to have a model of your body and the model of your environment and this is actually then already a form of self model which is part of the world model so the world model would be just be representation of the world or the relevant part of the world the the parts that are relevant to you and the your environment and your body is part of this world so it's the the self model is part of the world model but of course that's just a form of self model and it's quite primitive probably compared to the self models that we human beings have and one thing that distinguishes our self models is that we don't have only unconscious self self models but actually we have uh we also consciously experience the contents of our self models so this brings us on to to the notion of a phenomenal self model where phenomenal refers to phenomenal experience phenomenal Consciousness so if you have not just a um an unconscious self model but a phenomenal self model you actually experience your body as your own body and another form of selfhood that you then might experience is experiencing

your own

actions as your action so um this is

often called a sense of

agency um that you have over your action

and a sense of ownership that you have

over your body and both can

also um be

uh both can go get lost or be affected

in in various ways and so the these are

some Concepts that are relevant here

there's one thing that I wanted to add

or maybe an example that um Thomas also

likes to use I think is this robot

starfish that

um was developed by Josh Bongard and

colleagues and in order to make it

flexible and able it to adapt to changes

to the environment and also to its own

body so for instance one of its limbs

is um damaged or um is cut off um you

you want the robot to still be able to

control its body and control movements

and so it can be useful to have a model

of the body self model but of course in

that robot it's just a it's not a

phenomenal self model it's not a

conscious robot so that would be an

example of an

unconscious bodily self

model yeah I

let's start with the body I think that's

a good place to start because as you say

it's kind of the low hanging through the

lower the lowest level of what people

think about in terms of the phenomenal

self model and I think a potential point

of confusion for people is the fact that

the body has what I guess Marlo Ponte

call the jewel aspect which is both that it can be it's a lived body but it is also a body that can be observed and can be objectified and so I guess um the question here is why do we is there a difference between saying I am my body I live through my body or at or or versus I have a body and what are those different aspects of selfhood because on one part I feel like you're getting what we what we call perspectival I am my body I live through my body I live yeah I mean literally through it as a vehicle through which I can be in the world versus possession I am some in this case it seems like people would feel like a disembodied ego who has a body that they own is there a way of coalescing these kind of two canonical features of selfhood one being possession and one being perspectival or association with the lived body yeah that's a great question so I I think uh these different perspectives are to some extent compatible was it depends on on what you mean so one could for instance wonder whether a um animal non human animal which maybe does not have um a form of Consciousness that is comparable to to human conscious experience and that doesn't um reflect about its own body nevertheless lives through its body and so there there there's a live body but that doesn't mean that we don't have that we I would say it's just that we can also um have different forms of

experience or richer forms of bodily
experience and uh reflect about our own
experience in different
ways
and so having a body versus living
through a body so it it seems to me that
the Liv body is that that's
characteristic of
experiences or situations in which we're
not so much aware of having a body at
all so maybe when we're engaging in some
physical activity um playing sports and
just not not thinking about where our
limbs are but
just moving and
interacting and in the sense living
through the body I I don't know of that
captures what what is usually meant by
that expression yes I wonder whether I
mean it's a good question um we wrote
this paper not you and I me Carl L and
others wrote this paper on Flow States
uh and flow states are exactly this
you're playing sports and we tried to
ground it in a self-modeling
architecture and actually the revisions
have been really useful for that because
I've been able to get a little bit more
concrete on what I'm talking about and
the kind of dichotomy that we're saying
is present there I think I can say this
because hopefully it'll be published
well the prints out is um the
distinction really between a minimal
phenomenal self which is like these
really basic fundamental aspects of
selfhood mindness presentness
perspectival versus epistemic agency or

strong epistemic agency and by that I mean you know when we let's say plan or when I'm talking to you and I have to think about my next sentence there's a very strong sense of self-guiding that process where I can use let's say former resources former knowledge to apply to to uh deep temporal planning and we're saying that that's attenuated in flow and what we're also saying therefore is that metacognition because I think it's really important here to distinguish between pre-reflective self-awareness and reflective self-awareness and I'm treating reflective self-awareness as a metacognitive act your self is reflecting on itself because um reflective self-consciousness is itself a kind of a form of epistemic foraging that too is attenuated in flow States but I think the interesting question there is if I was in flow would I purely live through my body or could I be perceptually aware of having a body nonetheless could I have both um and these are very difficult questions to tease out because you know they're already done retrospectively and so you only get qualitative reports that's an interesting Dimension I had thought about I think okay so I I emailed you however you know a month ago two months ago and I said so I've been trying to get my I've been trying to think about how metzinger's phenomenal self model can be coales with active inference and this is the active inference inight podcast so I message you I said active

inference takes cognition to be
fundamentally
inferential are the processes within or
or the processes that generate the
phenomenal self model inferential are
the processes that are themselves being
modeled by the phenomenal self
inferential or both because I think van
you you would do a great service
everyone if you could explain that what
Thomas is talking about is kind of the
cognitive system recapitulating its own
data structures into phenomenal space
and I think that's an interesting idea
he talks about this in sort of his 2003
book and 2008 book and some other works
that the phenomenal self model is a in
in an implicitly reflective way is a as
you say a model of itself whether that
is a you know a giving relevant uh
features into Consciousness or Co grain
features into Consciousness whatever it
is but it's a reflective process where
the data structures in the body and
brain are being um transposed into
phenomenal space so maybe we can start
there I may have butchered it and I'm
sure Thomas is pulling his hair out um
but maybe we can start there and and
then feed in inference and see where it
might exist in within those
Cycles yeah so I I think there's
actually quite complex topic
because when we're talking about
self models it it it seems to
presuppose that there's already an
experienced eye right that I'm I'm I
have a body I am moving my hand and so

on but I don't think that's necessarily
already present in in any in in the most
basic forms of phenomenal self
models and what it's at least the way I
I think about it is that having a at
least a bodily self model is mainly
about just making a distinction
between one between inner and outer as
it were so I'm just trying to avoid the
word um self and and nonself so a
distinction between two types of
processes one is the environment and the
other
is here where I am and
and I I think that actually suffices for
having a a bodily self model or even a
phenomenal self model but of course when
we think about or our experience and the
way we the different forms of self that
we experience there's there's more to it
there there's something like an an eye
and maybe at this point it's useful to
um bring another aspect of the the self
model theory on the table which is the
phenomenal model of the intentionality
relation so quite

PM

exactly and so the idea is that there's
in in
a complex self model phenomenal self
model in which there's not just a
distinction between this belongs to me
this does not belong to me this is my
body this is not my body and so on well
this is
my my action or this is an uncon um
involuntary movement and so on in more
complex self

models you also have this this
perspectival which um and and
subjectivity um in the sense that that
you not just make the distinction
between what is self and nonself or what
is outside and what is inside but also
how you are related to the environment
how you are related to your your body
and how your your thoughts are related
to the world and so on so this brings a
reflexive component to it and makes it
more more complex um I find it a bit
difficult to put this into words so
Thomas could it is difficult I mean so I
can I can hopefully assist I have some
sort of quotes here um great so this is
the way that I've understood it at least
and so we'll start with the quote so
Thomas metzinger 2008 so the pap the ego
tunnel I think is 2009 so this is a
really actually very useful paper um he
wrote in 2008 something like empirical
approach to phenomenal self modeling
it's really it's actually very lucid he
says that
the subjectively experienced content of
the phenomenal self is the
representational content of a currently
active Dynamic data structure in the
system central nervous system and that
it's it's a representational entity
whose content is determined by the
system's very own properties so what I
mean by that is it's it's standing in
for something that is lower level to it
namely these data structures right and
we obviously here are taking as
axiomatic that there are

representational structures and maybe that's problematic and we can get there um and then the way that I've always read the PM is that the cognitive system can pay attention to the fact that it itself is directed at other things and therefore what happens is that it well I'm not going to say it makes the inference but it projects into phenomenal space this notion of itself and something else I guess my question here is that's taking as a presupposition that there is some division that the self model can model it can reflect on the fact that its own attent its own attention can be directed at so-called external worldly objects or things in the world model my question here would be how does the brain know what in the sense is external to it and what is internal to it in the sense that I can introspectively pay attention to parts of my own lived experience but not but have a sense of mindness and not feel that kind of Duality or that distinction but if I pay if in princip I pay attention to the chair next to me that feels you know like a separate object to me so if the brain is recapitulating some distinctions that are in its self model I guess my real question is are we not as you say already invoking an a priori distinction between the thing that's paying attention and the attended object yeah so if I understand

understand your question correctly um
then the answer would be yes and so what
you you um mentioned knowledge so how
does the brain know and of course
knowledge requires truth so you're
assuming that there is effect um to um
to the matter whether something is
really inside or outside or something is
really I or or mine or self or not self
and um I don't think we have to make
this assumption that there is that these
are truth claims and that the brain has
knowledge about this but it does have to
make some prior assumptions about that
and these can
be can be um
delusions so

um in in that sense I I think the
question how how does a brain know can
be dissolved but of of course then the
question is where where do these priors
come from and

[Music]

um why why are they relatively immune to
revision and what happens if they do
breakdown or the the sense of self
dissolves and so on so might it be
feasible to say that you know at the
very bottom level all we have is
attention in some way um recognize that
I almost well I did commit this kind of
dualistic fallacy when I said there is
the attender and the attended object
seems to me that actually that that
might be a an inference too far rather
there just is the attention onto these
kinds of objects in these data
structures again metzinger puts his far

be than I do he says that when he talks about the pmir the phenomenal model of the intentionality relation he says that you mentally simulate yourself as currently being directed at a Target object or goal State um and that's a really interesting notion right this mental simulation of yourself maybe he's committing a linguistic uh you know fallacy there that there's a self there in the first place um

so so we can pay attention let's say to the fact that we can be targeted at certain areas of our um Phenom of our data State space but then we can also pay attention to the fact that we're paying attention

and that seems to me I don't what you think about this proposal that seems to me to be maybe the heart of reflective self-consciousness that I can pay attention to the fact that there seems to be something that is paying attention to that which I'm that which is that where which is where attention is going to there's a kind of triple hierarchy um one does that sound like a feasible uh you know underlying explanation of metacognitive self-concept ization or reflective self-consciousness and

two how deep do those recursive patterns go what I mean by that is like how many times can I reflect on my own attentional processes and I know lar s Smith and others have been doing this in terms of So-Cal meta awareness but I'm

thinking more in terms of actually the underlying phenomenal reality of being a self is there a point in which the very notion just becomes uh you know irreconcilable with a certain degree of recursive operations yeah um great question so um regarding this this triple distinction I um I think that's already quite difficult to imagine so maybe we can uh just try to um clarify this and and just restate this again sure um together but before that there's something that um came to my mind because uh when we think about how how does a brain know um what what's inside um and what's in outside and but when it comes to I mean there there's some to to some extent there there might be truth to to these um to to claims about this and um so when it comes to attention we can just make a this really broad distinction between um attention attentional processes that are driven by external influences so when some Salient object or process process loud noise and the environment suddenly captures my intention and we can contrast that with cases in which I'm uh controlling my my attention and so the changes in attention are generated or caused by more caused by internal processes than by external processes and the the brain can make inferences about that and can be more or less accurate about whether some changes in attention have been more caused by external or internal

processes and it can make sense to model
attention to improve these inferences
and improve the control of attention and
so on maybe we can use this to just re
redescribe this triple distinction that
you mentioned because because I'm not
sure I I fully got what you okay yeah I
I I'm aware that there are lots of
layers to this um as is the problem when
we talk about hierarchies so I can
actually read I was thinking maybe it's
best for me to just read what I've so
this is this is so if people go and look
at that flow paper right now on the
pre-print server they won't see this
hopefully I'm not spoiling anyone's
party uh By the time this comes out hope
this is at the very end of a review
process so hopefully it's fine um anyway
people can shout at me if they want but
I wrote so it's my words um I said in
planning so we're talking about
epistemic agency in planning the
organism not only knows that it is
intentionally directed at something else
yielding a weak form of epistemic agency
and perspectival but it is also in the
act of knowing about its own knowledge
which encodes as M multiple
counterfactual representations of what
might unfold as a result of my actions
in the section of my current Eco Niche
that I seem to control in this way
planning involves strong epistemic
agency since the cognizer can
epistemically model itself as
epistemically agentive I.E it can come
to know its own capacity to control

itself and its econiche over protracted
period of time through stored knowledge
and then I go on to say that that
self-modeling process is transparent in
the sense that the agent often does not
know that it knows its intentional
objects rather there just is the sense
of epistemic agency which permeates a
pre-reflexive of self awareness this is
where the triple thing comes in when the
agent does enter a state of knowing the
fact that it knows its epistemic
capacities it evolves into a
phenomenologically metacognitive
metamodeling or perhaps meta epistemic
organism which reflects on its
being

yeah so that I I think the the issue
that I had when writing this was you
have to start from the original if
you're using the metzinger foundation
you have to start with the original
basis which is that the self model
itself is a simulation it's a
recapitulation of a data structure but
it's a recapitulation data structure
which can itself be uh reflected on and
that reflected that can also be
reflected on so that's where I'm
building in this nature of hierarchical
recursion how does that sound to your
ears and again what what what does it
mean for us to continue to go up these
recursive ranks like what happens for
example if I like is there a point where
I just really concretize my sense of
self or is it does it actually almost
become dispersive because all I realize

that's going on is that I'm paying attention and I'm just paying attention at different levels of the hierarchy so I'm kind of curious about what you think might be the phenomenological consequences of going up this hierarchy of reflective recursion yeah so I don't think I can give a full answer to this question because there might be multiple things going on so when I'm not just planning my my actions or planning how to how to interact with the environment or whatever but also reflecting upon my planning becoming aware that I'm planning I might be evaluating this I'm becoming aware of the fact that I'm uncertain about different options or I might become aware of the fact that I really want to do one thing and so this might then lead to further experience experience of of of certainty or maybe also um or or not uncertainty right or confidence that I'm making a good decision or doubtfulness about the decision and I think we can then repeat this process and reflect upon this already metacognitive process but the resulting experience that we have might not add too much so I don't know if it makes sense to say I'm I'm confident about the confidence that I have in mind my confidence and my confidence and so yes I mean it sounds

you know on one hand one could read that as a genuinely vertical recursion on one hand you can also read it as a kind of like the way people talk about the fact that a triple Bluff is the same as a single Bluff you know we might actually just turn our you know turn our heads onto our Tails so to speak and end up actually experiencing same it's a it's a curious inquiry um someone needs to model that maybe I need to model that um just um um what what I wanted to say is that I I think it makes a lot of sense for three levels or or um three recursions and then above and beyond that I I really don't have a clear intuition and it's probably an empirical question also to what extent it can improve control and um as I said just from the point point of view of experience we we we can mentally build further models further iterations but they might not have a real functional significance for the for the um lower order processes good okay we will park the self models I had to do it because it's been a bug bear of mind but we'll get Thomas on and we'll we can all chat out together and see where we end up um there's a broader question here about the so-call generative passage um which is this idea that uh is really nicely articulated in Maxwell Ramsey's 2022 Pap with and and colleagues of course um which is this idea that okay we kind of have two things that we are trying to balance one

is experiential reality lived experience
and the other are the in some sense the
tools of science that's one thing the
other thing could also be let's say
neurological
systems and um RAM and colleagues
present free forms of what they call De
generative passage or there's kind of
ways of tying those things together one
is ontological and so that's not
necessarily a subscription to
reductionism it's not necessarily saying
that phenomenal reality is
neural structures but that's one option
whether it's super that's that whole
conversation one is epistemological
which is that the kind of tools that we
use in our science um can be applied
fruitfully or can be used to describe
and explain phenomenal reality lived
experience and the last one is
methodological naturalization which I I
think I've been using a little bit more
in my own writing which doesn't really
have a commitment it's quite agnostic
about ontological epistemological claims
and just says well we can do some good
explanatory work by thinking about PP
schema or predicted processing or
whatever it is um but I'm not going to
say anything about whether Consciousness
is these uh these you know architectures
or neurons I was curious about I think I
think Thomas I I don't want to speak on
his behalf and you'll be able to say
better than I do I feel like he's quite
materialist or reductive about this um
and at least this is something that came

to my attention from Julian Kine when I spoke to Julian on this podcast um who was very keen to reject that reductionism and Forefront uh intentional experience and Forefront lived experience and say that that's irreducible so I was just curious about where you stand on on this generative passage and I guess the the sourest question is the ontological passage yeah um deep question so so first of all I I think different views on this all make sense and are useful and there there's a lot of uncertainty about how different levels of reality relate to each other and so to some extent we just have to follow some some assumptions that we make and and then there will be different views that are compatible with the models and data that we have so I'm not claiming that we can say well we we can obviously reduce um conscious experience to brain activity or something like that and it's to some extent also an open question for me what role computational models actually play in explaining conscious experience and what well do they play yeah so I I think there's when it so the maybe the simple answer or once simple answers to say well Consciousness is so puzzling it's so difficult to grasp its different features and talk about what we are experiencing from moment to moment it's difficult to put

that into
words and computational models can help
to make certain ideas a bit more
precise
and this might then just
be very on on a
very in in a very simple
way um a use of computational models not
that it's it's not even an explanation
it's just trying
to provide a better Des description of
that which we want to
explain what are the properties the
different features of conscious
experience that we need to explain and
computational models I think
can very legitimately be used
to provide such more detailed
descriptions and once you have that you
can of course then try to make
predictions
about the
implementation in the brain or make
predictions about how conscious
experience will change in certain
situations or you can compare
to
to um descriptions of conscious
experiences that we have to
phenomenological reports from persons in
Altered States of conscious
experience and so on and then at some
point you you may Wonder well maybe
these comp computation models provide
more than just a metaphor for this that
is useful to describe complex and
puzzling features of
experience maybe they also point to the

mechanisms and go some way towards explaining how conscious experience arises or why it has the features that it has even if we cannot maybe explain why there is conscious experience in the first place and to the extent that the computation models that are used are also neurobiologically at least not implausible and we might be able to find some empirical confirmation or to the extent that empirical neural biological and um data from cognitive cognitive Neuroscience are compatible with these models we might be inclined to make stronger claims about the relationship between conscious experience and computational models and neurobiological processes and of of course I'm so my I ideally we'd have very detailed computational models that can be used to describe predict and maybe even control many features of conscious experience and ideally we know how they might be implemented in the brain and therefore have some really have something like a computational um explanation of certain features of experience but I can also understand if some some people might be more hesitant in saying that though now that I'm thinking about it um I'm Tous actually I have to I guess I have to uh watch the episode with julian KY because

the long one yeah

okay yeah I'd be really interested in

what what exactly his his view on these

matters is yeah we go pretty deep

actually

um yes yeah he he was very much striking

down the line of the irreducibility of

conscious experience which I don't think

is necessarily that controversial um but

he was very

you know he was he was being stubborn

about that point I don't think he'd be

upset if I said that um yes yes okay

good um we can remain agnostic the hard

problem is the the hard problem and

unfortunately it's the most frequent

question that I ask uh and no one has

solved it for me yet and I don't think

they will so I'm G to ask a kind of what

Charmers might say is a less hard

problem or an easier problem um which is

about the features of Consciousness

especially as it's lived by us um is all

Consciousness self

consciousness um I would say no it's not

so I'm I'm I'm not an expert on this

this topic and I haven't read Thomas's

new book

NE but I um I intend to and hope

hopefully then I will know a bit more

about this but I take um reports from

people who have who claim to have

experienced States

of uh nonsubjective

experience seriously or think we we we

should not uh reject these

reports and I'm definitely open to the

possibility and also find the strategy

of trying to look at the simplest form
of conscious experience which may not be
a form of
self-consciousness I find that quite
promising yes so do I um does this map
on to something that I think you've
written about indeed you have um subject
unity and phenomenal Unity is there a
link there what well maybe we can start
with what's the difference between
subject unity and phenomenal
Unity yeah so that that's um
quite can can can become quite
complicated
it and so first of all subject Unity can
mean different things some people just
mean well different experiences that are
had by a subject are subject unified if
they're had by the same subject of
experience now what what is a subject of
experience and some people just identify
a subject with an organism so and in
that
sense subject Unity would
not
entail subjective experience or having a
having some form of
self-consciousness would just mean
subject Unity would then just mean that
there's a conscious organism that has
different experiences at the same time
so and these are all subject unified in
the sense that they are had by the same
subject but of course if you have a
formal self-consciousness
you might say that there's different
form of subject Unity maybe one in which
you're you you have

different

self-conscious experience or different
conscious experiences that are all forms
of self consciousness and somehow refer
to the same same

self something along these

lines yeah so what going back to the
kind of notion that all Consciousness is
self-consciousness goes back to actually
what Mr kin said um so he's got this
paper I think it's a 2018 paper I'll
double check that uh the no 2020 the
free energy in the self an ecological
inactive interpretation he Sayes very
clearly um he says in line with the
arguments of the phenomenologists I will
claim that every feeling must be felt by
someone it must have mindness built into
it if it is to feel a particular way
such a it's such a strong claim
and again I haven't read Thomas's new
book but I do kind of understand this
perspective because at least for example
let's talk about if if one was imagining
depersonalization it's a
depersonalization experience for who
right or if you think about someone
who's experiencing a psychedelic
experience it isn't so much that
perspectival itself is
eliminated it's more that one maybe has
this more Oceanic sense of self which
which is dispersive and that but that
still arguably is a
qualia for the self right or for the the
space of awareness that is inow with
perspectival so I'm just curious about
whether you can think again that this is

a tricky question it's a big question um
and there probably isn't in we can't
probably conceive of this at all but
sort of well I guess actually the
question I'll start with is when I learn
about Consciousness for example I
learned that there were three
fundamental tenants to a consciousness
experience qualia intentionality and
subjectivity this is what the British
schooling system teaches you it when I
hear sort of Julian speak about well
you're always going to have that perspec
tiveness or intentionality it's always
something for
you can we you know is there a way that
we could say okay some features at least
out of those three that I've just picked
up are more fundamental than others and
that we can have this very very minimal
conscious experience or minimal
phenomenal experience the MP another
acronym um
and what would you say are the defining
characteristics of a minimal phenomenal
experience
therefore yeah I can only say it's an
empirical question so we can speculate
about what the minimal form of
experience might be but it seems that
there's a lot going on in ordinary
waking experiences that can get lost in
Altered States Of Consciousness but what
exactly will remain I I I I don't know
so Thomas can say more about that is
there a problem
with relying on phenomenological reports
to inform such

questions yeah so to to some extent
there's of course the challenge that
these reports are given in retr respect
and oh how do you say retrospect yeah
just yeah yeah

yeah

and then so some people say well it's
it's a bit paradoxical if you say you
have had an experience in which you were
not present or something like
that but um I

I so and and one of of course the the
mere fact that you can remember such
experiences doesn't mean
you you you must have had a form of
self-consciousness or that there must
have been subjectivity in the experience
and also when people say well I realize
that I had not been there that there was
no sense of self or no no form of self
consciousness I realized that when
I when I transition transitioned into um
the ordinary form of conscious
experience so while as as it were the
the self was returning I realized hey
there there's

something additional
coming to my conscious experience so it
could not have been there
before yeah that's very interesting
that's really

interesting yeah I guess it makes me
think you know um to even have those
reports whether that's in the internal
speech or external reports requires
propositional language and in my
conceptualization of the scale of
phenomenal self- modeling propositional

language is super high up in the
epistemic agent it's the way that we
plan it's the way that we reflect on our
own being and so on and it sometimes
makes me think well is the lack of is
there a bias there when we don't have
the capacity for propositional claims
either internally or externally we're
biased to think that that constitute
some form of self Annihilation and the
reason why I think that is because like
testimonies for example about flow
States they'll say at least the
testimonies won't be as concrete as this
but the researchers interpretations
which are problematic will say the self
disappeared or there was there was a
reduced
self-awareness that's probably a weaker
form of that claim
well I think the all we're really coming
to here is that you just need to get a
lot more particular about what part of
the self model you're talking about and
how it's being attenuated or accentuated
um and actually I think hopefully for
the listeners what they're kind of
gathering is the self is not this kind
of single monolithic thing that we take
it to be and maybe that's the influence
of day cart and the tradition that
followed from him but rather it's a
complex
hierarchical well appearance in many
ways but also potentially underlying
data structure
that uh will oscillate and modulate and
change given different

contexts good um let's take that a little bit broader now into just yeah maybe just one one thing to to make it slightly more concrete so um what what is I I think very plausible is to assume that a sense of temporal self-location can get lost so you can have conscious experiences in which there's no time no temp experience you don't experience any now as it were or maybe only now but not as distinguished from future and past and then experiences in which you're not identifying yourself with some place in space so there's no spatial self location and if I understand Thomas correctly he would suggest that in such minimal phenomenal experience there certain some potential for certain control processes is is still experienced so what I mean maybe we can try to illustrate this with attention so you can either be in the process of controlling or changing your attention and have the experience of being in control have an an experience of attentional agency and you but you can also have a um a sense of control of the potential for changing your focus of attention so you you know now you're fixating focusing your attention on something or maybe on on my voice but you are aware of the fact that you could shift your focus of attention and this is so this this potential

aspect

might be something that in in one form

or the other could still be present in a

minimal phenomenal

experience and Thomas can

probably describe this more eloquently

and more detail it also makes me think

what are the prerequisites for

attention it makes me think that

something like temporality or just some

Dynamic needs to be there right because

how are you going to go from one thing

to the other without it being overtime

which also makes me think that it might

have to be also over space because well

I guess we also have covert you know

mental attention not covert attention

mental attention whether that's going

through some space is kind of depends on

how you define space that's a really

interesting question I guess yeah

there's a there's a nice link there you

could say okay I've got a function that

remains in the minimal phenomenal

experience but let me think also about

what the necessary features of the

experience or the world would have to be

for me to have that function um yeah it

makes me think of these um so-called

pure Consciousness events uh which I

came to know about through John BVI and

he sort of describes how in these pure

Consciousness events um so adjectival

qualia redness bliss sweetness

disappear but um we Call's adverbial

qualia remain and by that he means

nowness

heess presentness what you know

perspectival and maybe that's a nice
Road in but um yeah I will I probe that
with um with Mr matzinger your well I I
read I've read your papers before and I
read them again because I think they're
so wonderful and interesting you have
these two with call which I love
um and in one of them you talk about
sort of Life mind continuity which I've
also been getting really into because
I've been reading my Evan Thompson um
and in addition to sort of life and mind
distinction you also have a distin
between basic minds and non-basic Minds
which I think is important as a prior
thing to understand before we get into a
sort of Life mind continuity so at just
a very basic level what would you say is
the distinction between a basic mind and
a non Basic
mind um yeah so if I remember
correctly I I think the uh we we mainly
make this distinction in terms of
representation so a basic mind would be
one that does not have representations
and a non-basic mind is one that has
representations good and how does that
feed so how does that feed so um when EV
was talking about mind and life
continuity he was kind of seeking
underlying uh functions structures
processes that are involved both in the
constitution of mind and life so for
example autopoesis
self-organization um or what else would
he have had um operational closure these
are all things that were coming out of
sort of the dynamical systems theory and

Vera's work and now we've got Act of inference so I was wondering whether how does Act of inference because I think at one point you say in that paper um I won't missp speak I think I've got the quote here all systems possessing a mark of blanket have properties that are relevant for understanding the mind and Consciousness if such systems have mental properties then they can have them partly by virtue of possessing a Markov blanket and hence your marov and monism this is I think is this is the F this is a 2020 paper so not the life mind one but it is it calls to mind a sort of Life mind continuity because as inference postulates anything that exists has a mark of blankets and it's doing you know can be seen as if it's um parametrizing basian beliefs about the external state so everything has a markup blank anything that we can call a thing as a markup blanket and as you say anything having a markup blanket properties are relevant and I think you've been very careful with your words here for understanding mind why would that be the case and also how does that how is that the case without us resorting to a sort of pan psychism yeah that's a good question really fair question because I think what we say in that paper can easily be misunderstood and maybe yeah we should have found a better way of of conveying what where I I should have found a better way of conveying what I had in mind so the idea

is not that as soon as you have a mark
of blanket you have have a mind and and
it just becomes more complex
the idea is more that as soon as you
have the mark of blanket if you are self
organizing
system which conforms to the free energy
principle so there's a mark of
blanket
and you can then then you can interpret
internal
States as
encoding probabilistic beliefs about
external States given blanket States and
this is the feature that is also
relevant to understanding I would
claim the mind it does not mean that
once you have that you already have a
mind in an interesting sense but it's
just that if you have a
mind
then you can describe you can understand
I would
say mental processes in terms of minimiz
ing variation free
energy and then and and understand them
them in terms
of and in terms of computational
processes that involve these
probabilistic beliefs about external
States given blanket States and so the
the what what I really wanted to say is
that this is not something that comes at
at a later stage when we have really
complex system
but according to the FR energy principle
we already have that for very simple
self organizing systems and so to

contrast this with um an active
approaches that would
say well we have these simple systems
and they are still that they can be
quite autonomous and so on and
remarkable and so remarkable in fact
that even higher even nonbasic Minds in
in a different sense or maybe I
shouldn't say non Basic here but so even
more complex system most complex
organisms like us that have
Minds um they they many of their
properties can be explained many of
their mental properties can be
explained by reference to the same
Dynamics and features that are already
present in simple autonomous
systems and maybe you don't even need to
invoke representations to explain these
more complex
organisms and I'm suggesting well no
because you already have some form of as
if representationality or intentionality
and even simple self organizing
systems and this as if representation
ality can then um or you
can making some some further assumption
you can argue that in more complex
systems it's not just a useful way of
describing them as if they had
representations but they they actually
have
representations and so there there's
something in in simple self organizing
systems that already there uh that's
also there in in more complex organisms
that have a mind and Consciousness and
in order

to understand the
mind to the extent that the free energy
principle and active inference are
useful to understanding the mind we have
to involve we have to refer to these
internal States and how they encode
beliefs about external States given
blanket States and that's some something
we already have in very simple set of
organizing systems so that's just the
the idea which I think you if described
in the way in which we did can be really
misleading no I hope this also suggests
how it avoids pan
psychism not I'm happy to elaborate them
no I think so um and to be fair you make
it actually pretty clear um you say that
you sort of propose some rhetorical
questions does it follow that all
systems with a mark of blank have a mind
are such systems conscious this
formalism itself does not answer these
questions so I think you I think you're
fine I think uh maybe what people might
be slightly confused over is this notion
of an information geometry if they've
ever read this this paper so the way
that I understood it was you have sort
of two informational geometries let's
say two ways of uh describing a
um a system that has a mark of blanket
internal and external States right that
coupled system and you have the
so-called intrinsic geometry which
actually just describes the behavior of
the internal state if it you know um I
don't would that well I'll I'll ask
whether that includes the particular

States I the blanket States and then you have the so-called extrinsic geometry which are the beliefs that the internal States actually encode about the external States so that's a very sort of rough um draft that the distinction between the intrinsic geometry and the extrinsic geometry would be great if you could flesh that out a little bit more but how does that pertain here to um mind because I think you also make it quite clear that that property of the mark of blanketed system itself is relevant to the question of mind yes so it it is in a sense relevant but I would say in a rather trivial way so yeah I I think this can be really confusing and I'm not sure how useful it is

to even refer to information geometry to get at these more General conceptual points but so an information geometry is just you you have a space in which every Point corresponds to a probability distribution mhm and then you can um you can have different shapes and um manifolds in in this uh space so for example surfaces and compute distances between points distances between probability distributions and these distances will not be ukian distances so it's it's just um different form of um the different kind of space yeah and so every so the internal States if you interpret them as as U probabilistic beliefs about external States given blanket States you

can identify them with points in
the in an information geometry and then
the way in which these beliefs are
updated and which updated in which they
change can then be described as a
trajectory through that yeah um space
and you can AI and you also have a
different space of probability
distributions for the what one could
call the physical dynamics of the system
so if you're
not referring to probability
distributions encoded by internal
States but about the probabilities of
internal States or maybe an
paths then you you can also identify
them with points in in such a
space space of probability
distributions but
it's a different these are different
distributions and the thing just is that
you you can then describe trajectories
through both of these information
geometries and in in a way they are two
sides of two different sides of the same
coin so the idea is that you can
describe the physical dynamics of these
self-organizing
systems equivalently by describing them
in terms of the Dynamics the belief
Dynamics or basian
mechanics in terms of changes of
probability distributions encoded by
internal States rather than in terms of
distributions
Over States of the system yeah yeah okay
so I think the reason why this might be
a bit confusing for people is because

then people have heard that all that we're talking about when we talk about internal States is this parametrized if that's a word beliefs about external States that's all the internal states are right they just are distributions over external States okay so that that that might be a commitment anical commitment to say that but let's take that as a kind of starting point that's what we mean by internal States and then we have an intrinsic geometry which is mapping the evolution of those beliefs right um and that's over time but that given that those beliefs just are over the external States is there not an isomorphism between the intrinsic and the extrinsic in so far as when the intrinsic changes I.E when the path of the state of the internal States changes by necessity the extrinsic information geometry would have changed because the intrinsic the internal States just are the probabilistic distributions that the extrinsic geometry defines so for me I can't right now see how there's a I say they're isomorphic I almost think well I almost can't see how they're not one and the same but I might be missing something I probably am missing I'm always missing something no I I think you're on the right track so there yeah essentially just just two different ways of describing the system but okay so but do they so what is the utility of that I guess is the question

because to me it seems like
you you just a very philosopher you know
philosophy of science point the utility
of two different descriptions of the
same
phenomenon it has to have some utility
right like if we're talking about heat I
talk about how it feels to be hot or I
can talk about the velocity of particles
within a heat bar or within whatever but
that's useful because one I can attend
to phenomenological facts and the other
one I can attend to facts about
physics whereas this it just seems
like primary there are two distinct
descriptions but under the blanket
because the internal States really just
are probability distributions about
external States there's nothing over and
above their evolution beyond that which
can be described by the intrinsic
information geometry so is there
something more to these two types of
descriptions which make them distinct or
is it is it
just some semantics I I don't want to
say that because that always sounds very
derogatory and I don't mean to be
derogatory yeah so um on on on the one
hand what one could say yeah and to some
extent it's just semantics but that
doesn't mean that it's not useful to use
one instead of the other description so
think about describing a system in terms
of
representations so you you can of course
just
describe the vehicles without referring

to the content and so you're dispensing with the representational description and just focusing on the physical material realizers of the representations and that that's fine but um you typically get some benefit of going to a different level of description so when you're describing a system in terms of um representations you have you can refer to contents provided normative description so the they the system can have true or false representations or they can be more or less accurate yeah and so and and you also have that for uh the the Basin mechanics so it's in a way it's just a different way of describing the system and you don't need that you can describe the system you can stick to your physical purely physical description um and don't need to refer to probabilistic beliefs but this gives you some benefit so on on the one hand it reduces complexity because when you're referring to internal States as encoding probabilistic beliefs you're actually C graining the system and uh not just talking about the say the the um billions of of neurons but about neural populations that encode different beliefs and so you can um reduce the complexity of your

description but
still um given that if the free energy
principle is correct you um have an an
equivalent
description it's it can still be as
accurate or almost as accurate but
without having to deal with all the
unnecessary
details so that's one benefit that you
get just a reduction of
complexity yes I sense that that's a
reduction of complexity for the
scientist I guess really you very much
put it nicely in terms of the
distinction between the vehicle and the
content which certainly pertains when
we're talking about representational
systems I guess my point really here is
that I arguably think that there is no
vehicle the internal state is not a
vehicle of anything and I think this
really comes down to actually how you
describe it because the way you describe
it is that they encode probabilistic
belief whereas I think if you're
abandoning the vehicle you can just say
they are
prob um and if I think if you have the
vehicle but so sorry so then it seems
like you're dispensing with the physical
purely physical
description and just happy to to say
well we we only need the basil mechanics
right I I would just say that um what we
we are sometimes misled to think that
active inference entails a uh
representational approach
and yeah you so you can do the

descriptions in terms of the Bayesian mechanics that itself will map on to some physical organization

um but I don't think it necessarily dispenses with it it just says that when we as scientists try to carve up these spaces we are making well when we are making these very strong distinctions and this is something that I've been tackling with literally ever since I discovered active inference and learned about the free energy principle which is what is unique about the internal States right what what what is there something over and above the parametrization of beliefs about external states there in the internal states that make the thing the thing it I.E are is all that we are preferred States or are we a thing that is trying to get to preferred external States if you can see the difference there so that I think that again comes back whether we embody a generative model or have a generative model and I think I've been leaning in terms of the just in terms of the physics towards the idea that we embody a generative model all that means is that we are just a bundle of external States and you can describe the the the direction the information travel of those external States in terms of Bayesian inference but the internal external distinction is somewhat illusory because we just are the external States might sound very Buddhist actually but

um yeah I think maybe that's where we we
don't necessarily we're not disagreeing
but maybe where I don't really see the
strength of this this distinction
because I don't really see the strength
of the distinction between the internal
and the external and arguably I just see
it as a instrumental tool I don't know
if that answers your
question um yeah no no
um that's a good point also uh speaks to
what we talked about in the beginning
right to what
extent there's a fact to the matter
whether something is internal or
external yeah I just think a big mistake
that people well I think a big stumbling
block that people have although it may
also paradoxically be a very useful tool
for them is to think of active inference
and this is more like a meta
conversation about how we get educated
on these things think of active
inference as a theory that needs a
homunculus and I'd argue that it's not
but then you have to be very concrete on
the fact that free energy principle and
active inference is a Theory of
physics um and it's just
describing physical manifolds well not
even physical manifolds in this
statistical manifolds whether they're
physical mental so on and that I think
gets you away from the kind of more
hunzed predictive coding predictive
processing approach um and so again I
mean ultimately I think it comes down to
one's aims and I think as far as I'm

where are the math stands up on both fronts um and maybe it's a question I need to speak to KL about which is you know so the free energy principle rests upon the mark of blanket distinction why do we not just have one unified State space where things are tending well different things are tending towards different manifolds why do we need them to be tending towards those manifolds over the parametrization of external States and I think here the really important point is and we come back to over and over again on this podcast the free energy principle tells you what things look like they're doing not necessarily what they are doing um and maybe that deflates to use Carl's favorite word it deflates some of our worries about really making any strong ontological claim right it's just a descriptive thing then that makes me think well what's the uh what's the point but I don't think what's the point so I think it's important and I think you do get some you do get some good stuff out of it so yeah that there's just one thing I would like to add so um that I think there's one perspective from from which we one can argue that it's not completely arbitrary whether we say we interpret we describe internal States as probabilistic beliefs and identify them with with these beliefs and just focus on this or whether we also think about the the other

perspective other way of describing the
Dynamics of um the system because the
the same basian mechanics the same
dynamics of belief can be realized in
different ways and so you you can
simulate this on a computer or you can
implement it in a living
organism and I think when
we consider the possibility of
artificial Consciousness this actually
may make a
difference because if we're just
focusing on
the if we're just seeing internal States
as probabilistic beliefs and nothing
else then we cannot make this
distinction between different types
different kinds of
realizations
of variation of free energy
minimization and I think it's a an open
question and very important question to
what
extent there can be a conscious
experience in a computer in a digital
computer with a classical
architecture or
to what extent it needs a biological
body or maybe a robot that can interact
with the
environment
and for for this reason I I think it's
important to keep this to yeah not not
to um ignore this distinction between
two ways of looking at the system yeah
fair enough guess uh ultimately as well
I do think is important to say is that
it depends on how you define

beliefs um and that's something that we all like not me and you necessarily but all of us equivocate on the whole time because we're not clear where we're talking about folk psychology beliefs or basian beliefs and I guess when I'm talking about the internal States just being manifestations of external States I'm talking about basian beliefs in terms of the the beliefs that those internal states have when I'm talking about the rejection of the homuncular view of the internal States in many ways I'm re rejecting I'm not I'm not rejecting the basian beliefs that they have or embody or entail or so what I'm rejecting the folk psychology hunzed representational propositional beliefs that we talk about in philosophy so I think that's an important thing as well for people to bear in mind that there's often risk in Act of inference that we equivocate and talk past one another again not you and I generally but you know everyone because we don't get concrete on what we're talking about with respect to beliefs but that's a nice interesting um segwave because you wrote this paper in 2022 with Carl on AIs in computational Psychiatry and it actually takes us nicely full circles to Mr metzinger Professor metzinger me um because he actually was the first person I ever read in the ego tunnel talking quite explicitly about the effects of generating artificial intelligence like re and in a very philosophically robust way obviously people have been talking

about well what are the ethics in terms of job opportunities in terms of um you know trolley problems when my car might run over a grandmother or a child and so on but I think what Thomas said that was really impactful for me was we might be actually creating conscious life so it's not just the you know it's not just the the ethical effects on us it's actually the generation of sentience feeling Consciousness into the universe and uptick in that whereby those things start to actually have moral value which are arguably equivalent or perhaps given their greater complexity and depending on how you define your morality more important than us so you can throw out the window your kind of classic Sam Harris worry that AI is just going to stomp on us like we're ants right that's one ethical issue about us but also and again this is I don't mean that this should happen or will happen but arguably you know those AI creatures are going to have sentience and feelings and Consciousness just like us and we have to take their so you know rights or values if they have them if we give them that into account so it's not just a single sided relationship that's going to emerge I'm curious about what you think about that is it is it a concern that we really have to take seriously that if for example we end up making AGI and very importantly that AGI is conscious because I think there's a really important point there there's a difference here between intelligence and

Consciousness um what concerns do we have to have not only about ourselves but also about what we are bringing into the world

yeah good I think there are many different and to some extent related questions that are relevant here and one question is of course what what about the risk of actually creating artificial

Consciousness what consequences would that have

and should we do it or should should we refrain from even attempting to create artificial

Consciousness um there's things become more

complicated when we also consider the fact

that there will likely be a lot of uncertainty about

whether um a given artificial system is conscious or not so um Eric schwitz

Gable has done a lot of very relevant and interesting work on this and he describes the problem roughly as follows

I'll try to

summarize the main ideas but there's um people who are interested in this can

easily find his Publications and uh

podcast interviews with with ER Eric schitz Gable um so highly

recommend uh looking these things up and

um so the the basic problem is this um

there might be situations in which there are artificial IAL systems that may seem

to be conscious and we might in addition also have no way of being certain that

they are not conscious so there might be

no

um clear evidence as to whether they're conscious or not even from a scientific point of view um given that we don't have a consensus in Consciousness SS we don't really know what Consciousness is there's no um generally agreed upon Theory Of Consciousness and so on and many different metaphysical views about Consciousness that all have some currency so there's a lot of um uncertainty about which systems could be conscious and as artificial systems become more and more capable and have become have have more and more impressive cognitive capacities there will likely be points at which they will seem conscious and also fulfill some indicators of Consciousness that we might have but they might still be quite different from human beings conscious human beings or other conscious animals in in many respects so that there there will be a lot of uncertainty and there there two types of errors that one could make one is believing that such systems are conscious and maybe also believing that they therefore deserve moral status maybe because they can feel pain and pleasure um so assigning a moral status to such systems believing that they're conscious although in fact there's nothing going on they're completely unconscious and then there might be situations in which we have to make tradeoffs between the interests of human

beings conscious human beings and the
seemingly conscious artificial systems
which are in fact
unconscious so
just if we just think about uh um I mean
just just uh a slightly silly example
trolley case in which we we um can save
five artificial systems that we believe
are conscious or one conscious human
being and of course if these systems
actually are unconscious but we just
think well we
should U be well better safe than sorry
uh we we should not um we we should just
we on the side of being a bit too
liberal and we might uh say would be a
good idea to save the the artificial
systems or in in other situations in
general just um put more weight on their
interests and in
fact um they're they're completely
unconscious
and yeah so that that would be I think
yeah that's a terrifying idea yeah of
course the other case is that there are
artificial systems that don't seem to be
conscious and we firmly believe
or we we um maybe there's also some
uncertainty but um for whatever reason
we decide no um these the systems are
probably not conscious maybe because
they don't seem to be conscious but
actually they are conscious and then in
certain situations we
might without thinking much about it
sacrifice them or um destroy them or
whatever or and and in fact um maybe um
induce a lot um um a lot of suffering in

these artificial systems but without knowing that so these are two scenarios we I think would like to avoid and schwitz Gable certainly argues for this that we should avoid such situations namely by avoiding to create systems for which there's no certainty about whether they're conscious or not and so this is not not um it's a bit different from metzinger's position and um the the question of course is is this really feasible or is it realistic that we will in the future um only create systems that are either clearly non-conscious or that are clearly conscious and I think in order to get there it would be really useful um to have a handle on some necessary conditions for Consciousness that might not be fulfilled by certain classes of artificial systems and if for instance having a body is required for being conscious then we could rule out that Consciousness in a computer simulation so I'm not not claiming that I have a good argument for this but um it's I think it's would be extremely desirable to have a Justified account of necessary conditions for Consciousness that are not fulfilled by many artificial systems and so um that that's um the line of research that I'm trying to pursue of finding good reasons to believe that there are certain necessary

conditions for consciousness of course
the other approach which
um I I must say from
from um from a certain point of view
seems a lot more compelling is to say
well we don't really know what is
necessary for Consciousness or not we
cannot rule out that certain systems are
conscious if they don't fulfill certain
conditions but uh we
can look at theories of Consciousness we
can look at work on animal Consciousness
and derive indicators for
Consciousness and that's this um uh
preprint from last year by um Patrick
butlin Robert long and a whole bunch of
other authors Eric schitz Gale is also
one of the co-authors and they they do
exactly that they say well um under the
assumption of computation of
functionalism uh we can
derive certain number of indicators from
theories of Consciousness and can think
about how they could be implemented in
artificial systems in AIS we can also
assess to what extent they are already
fulfilled by current AI and but based on
this it's highly unlikely that current
AIS are conscious but if future AIS
fulfill many of these indicators it
becomes more likely that they're
conscious so each time a an artificial
system fulfills one of these indicators
the probability that it's conscious
Rises a little sure and of course that's
um very elegant approach because you
don't have to make any strong
metaphysical commitments you can just

say well we we don't really know what
Consciousness is but we can say what
features make it a bit more likely that
a system is
conscious and as soon as we have systems
that fulfill many of these indicators or
many markers of
Consciousness we might reasonably say
well it's highly likely that a system is
conscious and we
should
um assign moral status to the system but
of course then the question is what what
happens in the meantime when we have
many systems that fulfill some
indicators or many but
not maybe not so many indicators that we
can really be certain that these systems
are
conscious and how can we avoid these
situations yes it's fascinating it's
fascinating and we don't have time for
me to put on my philosophers skeptical
hat and
uh and point out all the problems with
trying to find behavioral indicators to
Consciousness which are so obvious that
I don't really need to say them but
again that's the philosophers point and
this more of a pragmatic point which is
we are probably going to have to make
that decision at some point and so why
is actually kind of interesting we're
saying up a sort of Bayesian process with
which to come up with a posterior that a
given object is conscious or not so it's
kind of fun that we're inverting the
Bayesian process to figure out um but as

you know your model is only as good as its priors and if those behavioral priors or behavioral indicators are wrong then you are really barking down the wrong tree you might be Bay optimal but your Bay optimal for dysfunctional purpose um cool that's really yeah that's fascinating I mean I'd love to go further but I'm aware that we are running a bit low on time so you mentioned this is kind of your main focus of research at the moment what else can people expect from you coming up next couple of months this year and also how can people get in touch with you if they want to ask you questions or chat yeah sure um so I'm I have a blue sky

account and um I I think my blue sky handle is va. Visa probably should check but um

I also have a Twitter account I don't really understand what it is I like Twitter

but yeah so so I also have a Twitter account and then my handle is w a w s e I think

excellent and um and what can people look forward to reading or or hearing from you about um and also maybe plug the workshop that you're at now because that's a really cool um thing that you guys are doing yeah so I'm currently at um the ik in gunner in Germany um ik is um the interdisciplinary College it's a spring school for cognitive science and it's an entire week with a lot of fascinating courses MH from AI to

neuroscience neurotechnology um we also have some more philosophical courses uh some more theoretical some more practical it's a really great mixture and there many fascinating people over here that you can talk to so I highly recommend coming to the institute and Guna and um maybe do you have show notes or something like that for the podcast maybe you can provide link to the website is everything goes in the description I'll put your blue sky in the description as well I'll make sure I do that thank you cool and also maybe maybe one thing what what people can expect um so I signed a contract with MIT press um for a book on artificial Consciousness so working on and also um I think I should remind myself that I should be busy working on the manuscript fantastic well congratulations is that uh do people know this already or is this a big reveal have we done have we done an exclusive break on active inference insights or do people is this do people know about this book already um I think most people don't know about it I've just told a few persons about it but on the other hand most people don't know me so yeah true no one knows us well yeah I forget that I was just trying to sort of embody a you know speculate you know speculative Paparazzi esque person I it very well van this was super fun um I feel like there's still a lot to get

into um so maybe when Thomas comes it
would be great to have you on even if
you know partially for a bit so we can
get deeper into these ideas and um thank
you for plating me and listening to my
ramblings about recursive properties and
operations it's no thank you thank you
very much for for having me and then
this was really a pleasure so um I think
it we had a nice flow in the
conversation we did we did all right
thank you everyone for watching as well
take care